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REMAGNETIZATION BY BASINAL FLUIDS: TESTING THE HYPOTHESIS IN THE VIOLA LIMESTONE, SOUTHERN OKLAHOMA

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Abstract. Migration of orogenic or basinal fluids is a recently invoked mechanism to explain the widespread presence of late Paleozoic secondary magnetizations in the rocks of North America. Paleomagnetic and geochemical results from the Ordovician Viola Limestone in southern Oklahoma are evaluated to assess the role of basinal fluids in leading to secondary magnetizations in the unit. The Viola Limestone contains what we interpret to be a pervasive Pennsylvanian synfolding magnetization residing in magnetite and a localized Permian magnetization which resides in hematite and occurs in alteration zones around mineralized veins. Both secondary magnetizations are interpreted as chemical remanent magnetizations (CRM) based on low burial temperatures and the presence of authigenic magnetic phases. The relative proportion of the Permian CRM in hematite gradually decreases whereas the magnetite CRM increases with distance from the veins. Fluid inclusion and Sr isotope studies indicate that the vein mineralization (calcite with Mississippi-Valley-type oxides and sulfides) precipitated from basinal fluids which were warm, saline, and radiogenic. The radiogenic $^{87}Sr/^{86}Sr$ values of the limestones in the alteration zones, and the fact that there is more significant alteration closer to the veins suggests that the basinal fluids were also responsible for alteration in the limestones. The coincidence of the geochemical and remagnetization trends suggests that the Permian CRM dates the migration of basinal fluids through the veins. Geochemical results from the Viola Limestone containing the pervasive CRM indicate that it is relatively unaltered with no evidence for basinal fluids. The lack of evidence for basinal fluids suggests that other mechanisms for the origin of the pervasive CRM need to be tested. The results of this study indicate that flow of basinal fluids was focused in veins and only locally altered the host limestone.

Introduction

Secondary magnetizations are common in sedimentary rocks, and a number of mechanisms have been invoked to explain these magnetizations (as recently summarized by Elmore and McCabe [1991]). Many secondary magnetizations are interpreted to be chemical in origin and are evidence of one or more diagenetic events. If a direct connection between a secondary magnetization and the diagenetic event can be established, then paleomagnetic data can provide important information on the timing and nature of the event.

The role of fluids in general, and particularly orogenic fluids expelled as a result of tectonism, has recently received considerable attention as a possible agent of remagnetization [e.g., Oliver, 1986; Miller and Kent, 1988; McCabe et al., 1989; McWhinnie et al., 1990; Pan et al., 1990]. The evidence for the postulated connection includes a spatial association between many secondary magnetizations and orogenic belts as well as a temporal association with orogeny [e.g., McCabe and Elmore, 1989]. This particularly applies to many late Paleozoic secondary magnetizations, which, along with other phenomena including widespread potassium alteration in the late Paleozoic [e.g., Hearn and Sutter, 1985; Hearn et al., 1987; Elliot and Aronson, 1987; Hay et al., 1988], high-temperature alteration of coal [Daniels et al., 1990], and trapping of hot fluids in fluid inclusions [e.g., Leach and Rowan, 1986; Dorobek, 1989], are inferred to be genetically related to the expulsion of hot brines from the Appalachian and Ouachita fold-thrust belts [e.g., Oliver, 1986, 1992; Bethke and Marshak, 1990].

Although there is a clear association between orogeny and remagnetization, the evidence for a connection with orogenic fluids is circumstantial. For example, the chemical composition and origin of the fluids that presumably cause the secondary magnetizations are usually not well constrained. If paleomagnetism is to provide a truly useful record of the presumed chemical event, then tests for direct links between secondary magnetizations and the responsible fluids are needed.

Geochemical studies, in conjunction with paleomagnetic and rock magnetic investigations, can provide important constraints on the origin of chemical magnetizations. In this study, paleomagnetic and geochemical data are used to determine the origin of two secondary magnetizations in the Viola Limestone (Late Ordovician) in southern Oklahoma. One localized magnetization occurs around veins which contain minerals precipitated from basinal fluids. A

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second pervasive magnetization resides in magnetite and is similar to many other secondary magnetizations in carbonate rocks, the origin of which are controversial [e.g., McCabe and Elmore, 1989]. The results of the study have implications for dating migration of basinal fluids and understanding remagnetization mechanisms.

Geologic Setting

The Viola Limestone in southern Oklahoma is up to 300 m thick and consists primarily of gray fossiliferous mudstones, wackestones, and packstones. The unit contains carbonate laminites with abundant chert lenses at the base and grades up to grainstones at the top. This sequence records deposition on a deep to shallow carbonate ramp within and peripheral to the Southern Oklahoma Aulacogen [Galvin, 1983]. The Viola is separated from the overlying Sylvan Shale by a Late Ordovician disconformity. An estimate of the burial depth for the Viola in the Late Mississippian is approximately 3 km. During the dismemberment of the Southern Oklahoma Aulacogen in the Pennsylvanian the Viola was deformed and uplifted [Ham, 1978; Johnson et al., 1989].

Primary porosities in the Viola limestones, particularly the mudstones and wackestones, were low. Any porosity present was apparently occluded by an early phase of meteoric cementation [Grammer, 1985]. The Viola produces oil in some areas, and the porosity is a result of fracturing caused by deformation [Allen, 1983].

Samples were collected from several locations in southern Oklahoma (Figure 1). Most samples were collected from an anticline-syncline couplet that is well exposed in the Sooner Rock and Sand (SRS) quarry in the northern flank of the Arbuckle Mountains (Figure 2). The axes of the folds trend NNW-SSE, and both plunge gently to the south. The eastern limb of the anticline is fractured. The fractures have two general orientations. One set strikes approximately N25°W, dips 50° to the southwest, and is apparently axial planar to the folds [Bagley et al., 1992]. The second set is approximately normal to the folds (N55°E to N60°E) and is likely to represent extension fractures [Wiltse, 1978]. Many of the fractures are mineralized with calcite and other phases (Figure 3). The host Viola around these mineralized fractures is also reddened, altered, and in some cases bleached. Some of the fractures are pipelike and others are conduits for tar seeps. One highly fractured, brecciated area contains slickensides and is interpreted to be the surface extension of a fault the trends NNW-SSE and, on the basis of subsurface data, dips to the west [Wiltse, 1978]. Examination of cuttings from the fault zone in nearby wells indicates the presence of mineralization.

In the western part of the anticline-syncline couplet the rocks contain small millimeter-scale veins filled with calcite. These veins contain only calcite mineralization, and there is no alteration around them. Many veins contain calcite which is twinned and deformed.

Within and around the SRS quarry are a number of oil wells (Southwest Davis Oil field) that produce from 1000-1550 m depth. The quarry also lies approximately 3 km NE of the Southwest Davis Zinc deposit, where the lower Ordovician Butterly Dolomite contains sphalerite [Fay, 1981]. Sphalerite also occurs within a broad region that includes localities several kilometers to the north and east of the SRS quarry.

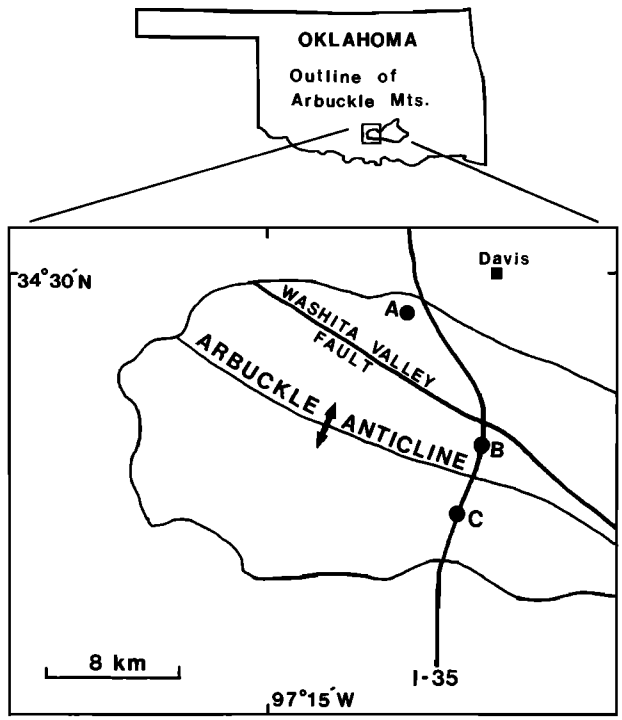


Fig. 1. Location map of study area. Samples were collected from the Sooner Rock and Sand (SRS) quarry (A) and from roadcuts along U.S. Interstate 35 (B and C).

Samples were also collected from the Viola Limestone on the flanks of the Arbuckle Anticline (Figure 1). The south flank dips uniformly to the south whereas the north flank is complexly folded, faulted, and overturned. The rocks are similar to the nonmineralized zone in the SRS quarry. Peck et al. [1984] presented preliminary paleomagnetic and rock magnetic results from the Viola at the Arbuckle Anticline which indicated the magnetization resided in magnetite. Directions for the magnetization, based predominantly on alternating field demagnetization, were southeasterly with moderate to shallow inclinations.

Vitrinite reflectance values from the Woodford Shale in the northern Arbuckle Mountains are low and indicate burial temperatures below 100°C [Cardott et al., 1990]. The Woodford is approximately 150 m above the Viola in the study area, and we have no independent evidence to lead us to believe that burial temperatures in the Viola exceeded 100°C.

Methods

Over 160 samples from 25 sites were collected from the Viola in the SRS quarry and from the Arbuckle Anticline. Sites sampled on the flanks of the Arbuckle Anticline in the study by Peck

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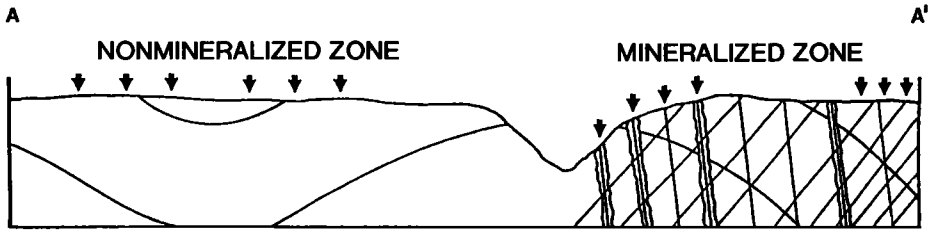


Fig. 2. Sketch map of a cross section through the part of the Sooner Rock and Sand quarry where the samples were collected. The two zones sampled are noted, and the arrows locate sample sites. A is to the west; A' is to the east.



Fig. 3. Photograph of a mineralized fracture from the mineralized zone in the Sooner Rock and Sand quarry. Note that the pipe is lined with calcite. Drill holes are 2.54 cm in diameter.

et al. [1984] were resampled because we were concerned that alternating field demagnetization may not have fully isolated the characteristic magnetization [e.g., McCabe et al., 1984].

The magnetization was measured using a Sct cryogenic magnetometer located in a magnetically shielded room. Most specimens were subjected to stepwise thermal demagnetization using a Schonstedt TSD-1 oven. Some specimens were subjected to alternating field (AF) demagnetization in a two-axis tumbling unit inside a variable field electromagnet. The demagnetization results were plotted on orthogonal projections, and principal component directions [Kirschvink, 1980] were calculated.

Isothermal remanent magnetization (IRM) acquisition and decay measurements were performed to aid in the identification of the magnetic mineralogy. Specimens were subjected in a stepwise fashion to an increasing IRM up to 0.8 T. The specimens were thermally demagnetized in 13 or more steps. Some specimens were subjected to a composite IRM (0.8 T in one direction and 0.2 T in the opposite direction [Channell et al., 1982]) and thermally demagnetized.

Petrographic studies (both transmitted and reflected light) were conducted on thin sections of chips cut from the paleomagnetic samples. Magnetic extracts from the Viola were also obtained by crushing bulk samples into small chips in a clean environment and isolating the magnetic chips with a strong magnet. Although time consuming, this approach allowed the magnetic phases to be examined in situ. The chips were examined with a scanning electron microscope (SEM) that has energy dispersive (EDA) capability. Polished thin sections were also made of several magnetic chips and were examined using a microprobe.

Geochemical data were obtained from samples of the calcites in the veins and of the host Viola Limestone. Isotopic analyses were performed on samples of lime mud, the dominant lithology sampled in the paleomagnetic study of the host Viola Limestone. Sampling of fossils was avoided and the millimeter-scale veins in the host Viola Limestone were microsampled. The large veins and associated altered rock were also sampled. Details of the procedures used in the isotopic (Sr, C, and O) and trace element investigation are described by Gao et al.

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[1992]. Details of the procedures used in a fluid inclusion study of the vein calcites are described by London et al. (manuscript in preparation, 1993). Solid inclusions in the calcites were identified using the microprobe. Xray diffraction (XRD) analysis was also performed on some samples using equipment at the University of Oklahoma. Insoluble residues were prepared by dissolving representative samples in buffered acetic acid.

Organic geochemical studies, including determination of total organic carbon (TOC) and Rock-Eval pyrolysis, were also conducted on some samples from the Viola Limestone. These analyses were performed in the Organic Geochemistry Laboratory at the University of Oklahoma.

Geology and Geochemistry of the Viola Limestone

Introduction

Details of the geochemical and petrographic studies of the veins and the host Viola Limestone are presented by London et al. [1990; also manuscript in preparation, 1993] and Gao et al. [1992] and will be summarized here.

There are two general types of veins, one with asphalt and another devoid of asphalt. Most veins also contain at least two generations of calcite deposition: a basal scalenodedral calcite and a rhombic overgrowth. Asphalt-absent veins contain pyrite at the contact between the calcite forms with goethite (Figure 4*a*), hematite, pyrite (Figures 4*b* and 4*d*), minor sphalerite, and minor chalcopyrite along growth surfaces in the rhombic overgrowths. Asphalt-bearing veins contain no solid inclusions within the calcites, although pyrite and sphalerite (Figure 4*c*) are present at surface contacts between the two calcite forms.

The basal scalenohedra in both types of veins contain primary fluid inclusions (Type I; Figure 4*e* and Table 1) which indicate deposition at approximately 70°C from a brine containing 15-20 wt % salts with equivalent wt % $CaCl_2/[NaCl + CaCl_2] \approx 0.75$. In asphalt-absent veins the subsequent deposition of rhombic calcite overgrowths occurred at 50°C and primarily from a different brine dominated by monovalent cation solutes (Type II inclusions; Table 1). Type II inclusions, which are primary inclusions along growth planes, occur with other inclusions (Type III, Figure 4*f* and Table 1) that are

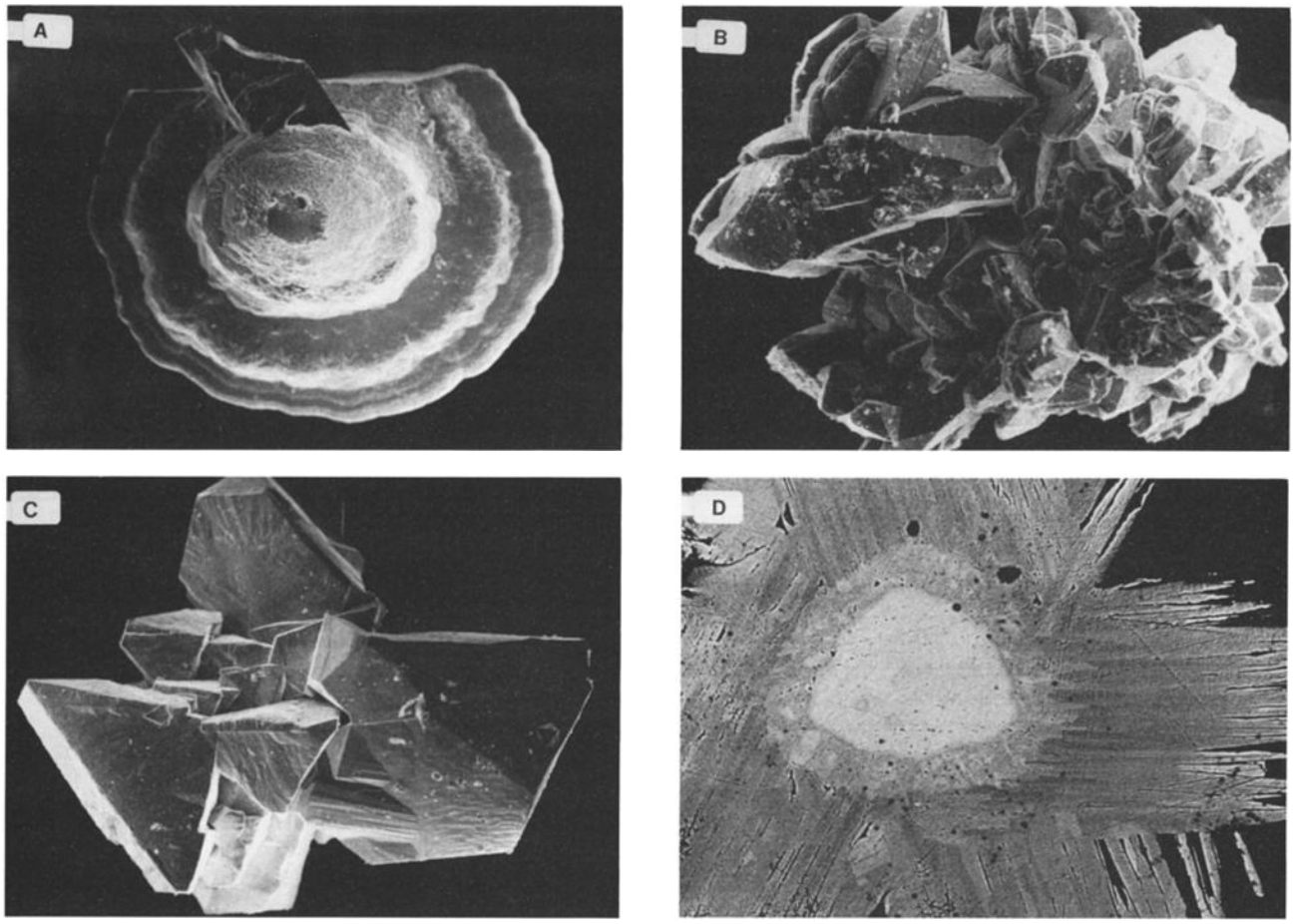


Fig. 4. Accessory minerals and fluid inclusions associated with vein calcite. (*a*) Scanning electron microscope (SEM) image of a mushroom-shaped aggregate of goethite, dissolved out of calcite from an asphalt-absent vein. Each mushroom is composed of an axially radial growth spindle of tiny acicular goethite needles, and a spherically radial growth form that defines the cap. The growth direction was from base of spindle toward cap. (*b*) SEM image of highly distorted octahedral pyrite crystals that are high in Ni and Co (several weight percent of each). Some pyrite

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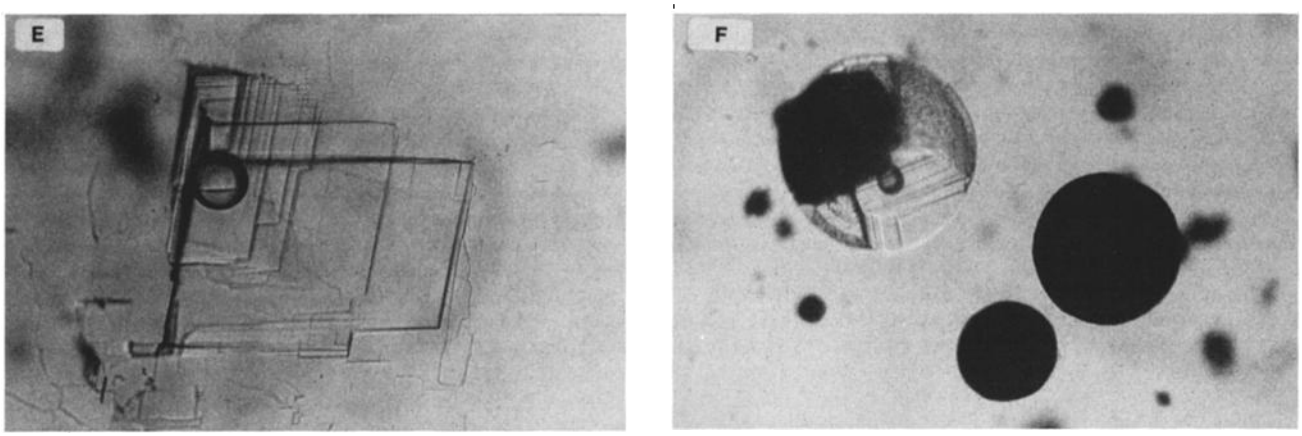


Fig. 4 (continued)

may be after marcasite. Minor chalcopyrite is present in this and other aggregates. The sample is from a contact between scalenohedral and rhombic calcite overgrowth in an asphalt-absent vein. (*c*) SEM image of sphalerite from the tips of scalenohedral calcite in asphalt-bearing veins. This sphalerite is typical of the grain size, it is high in Fe, and it contains local growth zones rich in Hg. (*d*) Backscattered electron (BSE) image of a composite pyrite grain, dissolved out of a rhombic calcite overgrowth in an asphalt-absent vein. The high BSE sector at the core consists of a pyrite solid solution with approximately 6 wt % each of Ni and Co. The lamellae with low BSE signal intensity form a skeletal octahedral overgrowth of pyrite near end-member composition. The sharp compositional definition is taken as evidence for pyrite growth from two fluid phases: one rich in base-metal cations, such as the early Type I basinal fluid, and a second influx of fluid, perhaps corresponding to the Type II fluid inclusions, of different chemistry. Mixing between Type I and Type II fluids is inferred to have caused rapid, disequilibrium deposition of sulfides, as evidenced by finely zoned crystals of pyrite and sphalerite, and skeletal growth forms such as these octahedra. (*e*) Photomicrograph of a Type I aqueous liquid-vapor inclusion (at 25°C) in scalenohedral calcite. The inclusions is approximately 75 μm across. (*f*) Photomicrograph of a Type II-III aqueous liquid-vapor inclusion (at 25°C) in rhombohedral calcite from an asphalt-absent vein. The spherical form and rough-textured walls are typical of these inclusions. The inclusions occur in the growth planes that contain radial goethite aggregates (spheres and mushrooms), pyrite, and rarely sphalerite and chalcopyrite (i.e., the associated opaque phases in this photo). The inclusion is approximately 50 μm across.

TABLE 1. Summary of Microthermometric Data

	X	S.D.	*N*
	Type I Inclusions, Scalenohedra*		
Te	-54°C	8°C	110
Tm	-20°C	3°C	110
Th	+71°C	7°C	22
	Type II Monovalent Inclusions, Rhombs†		
Te	-26°C	2°C	42
Tm	-8°C	4°C	42
	Type III Mixed Inclusions, Rhombs†		
Te	-42°C	7°C	155
Tm	-8°C	4°C	155
	Type II and III Combined, Rhombs		
Th	+49°C	5°C	30

Te is eutectic temperature; Tm is ice disappearance temperature; Th is temperature of homogenization of vapor to liquid. S.D. is standard deviation. *N* is number of samples.

*From veins with and without asphalt.

†Present only in veins without asphalt.

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indistinguishable (in size, shape, distribution, bulk salinity, and homogenization temperatures) except in their generally lower and more variable eutectic temperatures, which range between Type I and Type II inclusions. These are interpreted to have resulted from mixing of the Type I and Type II fluids. Most calcite samples also contain a few secondary inclusions along healed cracks, many of which lack vapor bubbles at 25°C, that appear to represent fresh near-surface waters. Other than rare secondary inclusions, there are no fluid inclusions in rhombic calcite from asphalt-bearing veins.

The calcites in the veins have low Sr (Figure 5*a*) but high Fe and Mn concentrations (Figure 5*b*). Isotope analysis (Figures 5*a* and 5*c*) of the calcites indicates radiogenic $^{87}Sr/^{86}Sr$ ratios (0.70792-0.70880), light $\delta^{13}C$ values (-2.5 to -8.0‰, PDB [PeeDee Belemnite]), and relatively heavy $\delta^{18}O$ values (-3.7 to -4.8‰). The isotopic values from the different calcite forms are similar, although the rhombic overgrowths in the asphalt-absent veins contain relatively heavy $\delta^{13}C$ values and higher concentrations of Fe, Mn, and Sr compared to the other calcite forms [Gao et al., 1992].

In contrast, the results from the host Viola (and associated small veins) are different (Figures 5*a*-5*c*). The Viola has high Sr but low Fe and Mn concentrations. Sr isotope ratios are lower (0.70779 to 0.70794) and the $\delta^{13}C$ values are heavier (-2.2 to -1.7‰) than they are in the vein calcites. The $\delta^{18}O$ values range from -3.1 to -5.0‰ (PDB).

TOC values from the Viola Limestone at several localities (e.g., south flank of Arbuckle Anticline) range from 0.06% to 0.54%. Rock-Eval pyrolysis was performed to determine the relative maturity of the Viola. The resultant s1/s1+s2 ratios range from 0.06 to 0.21, and Tmax ranges from 434°C to 442°C.

Petrographic investigations reveal that the Viola contains small amounts of detrital quartz (1-10%), potassium feldspar (<1%), and plagioclase feldspar (<<1%) [Sedigi, 1984]. K_2O weight percentages for the Viola in the Arbuckle Mountains average 0.09% and range up to 0.19% [Sedigi, 1984]. XRD analysis of unoriented samples indicate predominantly calcite with minor amounts of quartz. Authigenic pyrite is abundant in many samples, and a number of different forms are present, including framboids. The insoluble residue content of bulk limestone samples was less than 1.5%. Thin shaley lenses interbedded with the limestones, however, are predominantly composed of clay. XRD analysis of oriented insoluble residues indicated illite as the only clay mineral present in the limestone.

Examination of magnetic chips from the Viola Limestone using the SEM indicate the present of spherical iron-rich phases in situ within the limestone (Figure 6*a*). In addition to Fe, most the spheres found in situ also contain Ca and Si, and some contain Ti, K, and Al. Some samples contain replaced or partially replaced pyrite framboids (Figure 6*b*).

Around most veins (those without hydrocarbons and few with hydrocarbons) there is an alteration zone characterized by a yellow to reddish stain. The alteration zone around some of the veins with hydrocarbons is bleached

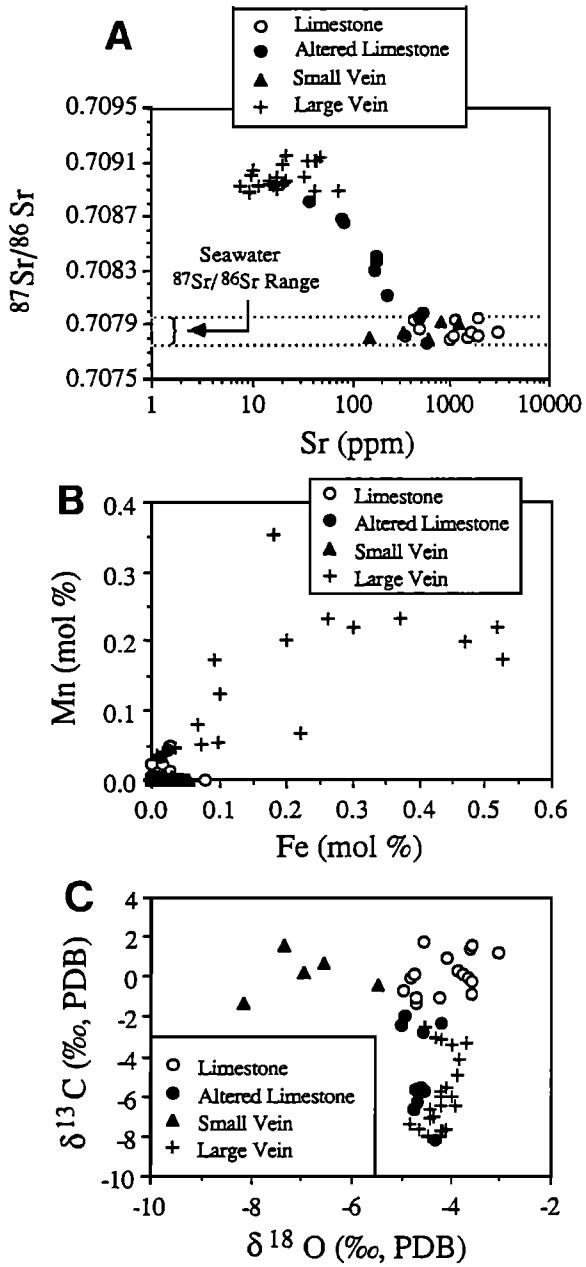


Fig. 5. Geochemical results from the Viola limestone. (*a*) Sr versus $^{87}Sr/^{86}Sr$ plot, illustrating that limestone and small calcite veins have $^{87}Sr/^{86}Sr$ ratios similar to late Middle and Late Ordovician seawater (about 0.70775 to 0.70795 [Burke et al., 1982]), whereas altered limestone and the large vein calcites are radiogenic. (*b*) Fe versus Mn plot showing that the large calcite veins have high Fe and Mn concentrations relative to the Viola limestone and the small calcite veins. (*c*) $\delta^{13}C$ versus $\delta^{18}O$ plot, highlighting that (1) altered limestone has lighter $\delta^{13}O$ but similar $\delta^{18}O$ values to host limestone, (2) the large calcite veins have heavier $\delta^{13}C$ and lighter $\delta^{18}O$ values than the small calcite veins, and (3) the $\delta^{13}C$ values of the small veins are similar to Viola limestone, whereas the $\delta^{13}C$ values of the large veins are consistent with those of altered limestone.

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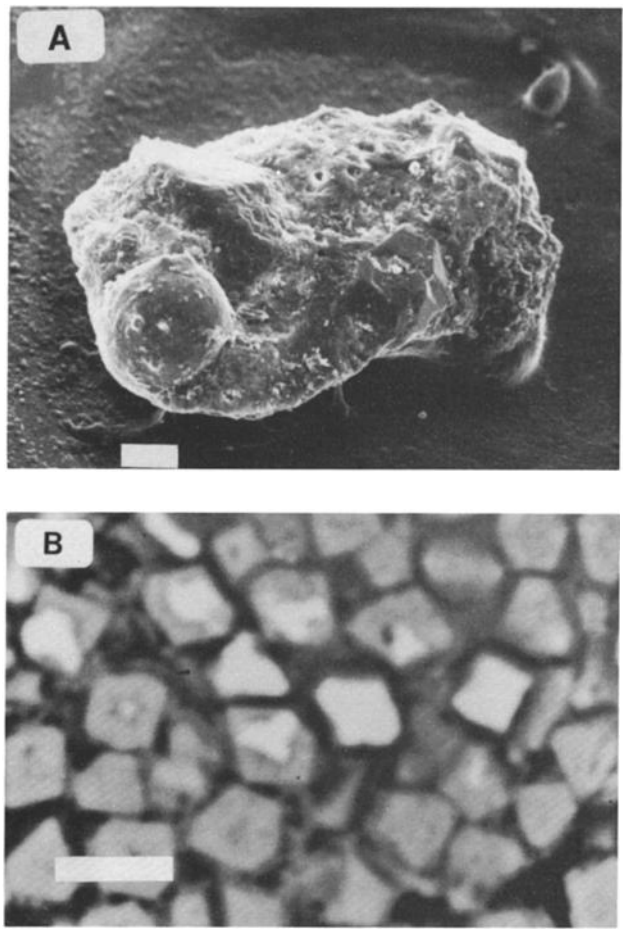


Fig. 6. Scanning electron microscope photomicrographs of phases from the nonmineralized zone. (*a*) Iron-rich sphere within the calcite matrix of the limestone. Scale bar is 50 μm. (*b*) Backscatter photomicrograph of oxide (darker material) replacing pyrite (lighter material) in a pyrite framboid. On the basis of microprobe analysis we could identify the pyrite, but we could not determine the oxide phase. Scale bar is 3 μm.

rather than reddened. The limestone in the alteration zones is recrystallized. Small calcite veins extend from the large veins into the limestone (Figure 7*a*). A secondary fine-grained calcite commonly occurs in vugs and small fractures at the contact between the major veins and the limestone. This material is usually stained red or yellow, although around some veins with hydrocarbons, it is gray-white. In some cases, calcite veins cut through this material.

Authigenic hematite is abundant in the reddish alteration zones, particularly near the contact with the veins. A fine-grained red hematite pigment stains the host limestones. Authigenic specular hematite is also present, and some hematite has pseudomorphically replaced pyrite (Figure 7*b*).

Sr concentrations and Sr isotope ratios from the alteration zones are intermediate between the host limestone and the large calcite veins (Figure 5*a*). The $\delta^{13}C$ values are also light (-2.0 to -8.2‰, PDB). With increasing distance from the veins, the Sr isotope ratios and $\delta^{13}C$ values decrease whereas the Sr concentrations increase (Figure 8). The results from the reddish and bleached alteration zones are similar.

Interpretation

The geochemical results from the nonmineralized Viola Limestone suggest that it is relatively unaltered. For example, the $\delta^{13}C$ values from the Viola Limestone are similar to primary $\delta^{13}C$ values of Middle and Late Ordovician carbonates [Lindh, 1984; Popp et al.,1986; Veizer et al., 1986]. This suggests that the $\delta^{13}C$ values probably represent unaltered values of the original Viola sediment. The $^{87}Sr/^{86}Sr$ ratios of the Viola Limestone are consistent with worldwide seawater ratios for the late Middle to Late Ordovician time [Burke et al., 1982] and therefore probably represent the $^{87}Sr/^{86}Sr$ ratios of the seawater in which the Viola was deposited. The Sr values in the Viola Limestone are high and suggest that the original sediments were aragonitic in composition. The Sr values, however, are low compared to modern

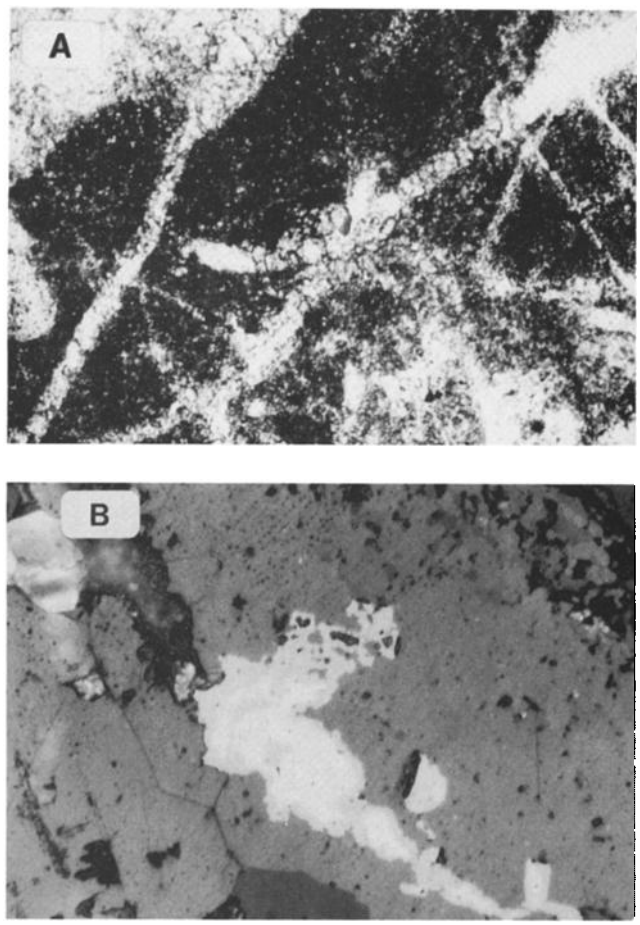


Fig. 7. Petrographic characteristics of the alteration zones around the fractures. (*a*) Small calcite veins from an alteration zone near a large vein (transmitted light). Scale bar is 300 μm. (*b*) Authigenic hematite from the altered limestone near a vein (reflected light). This hematite may have pseudomorphically replaced pyrite. Scale bar is 30 μm.

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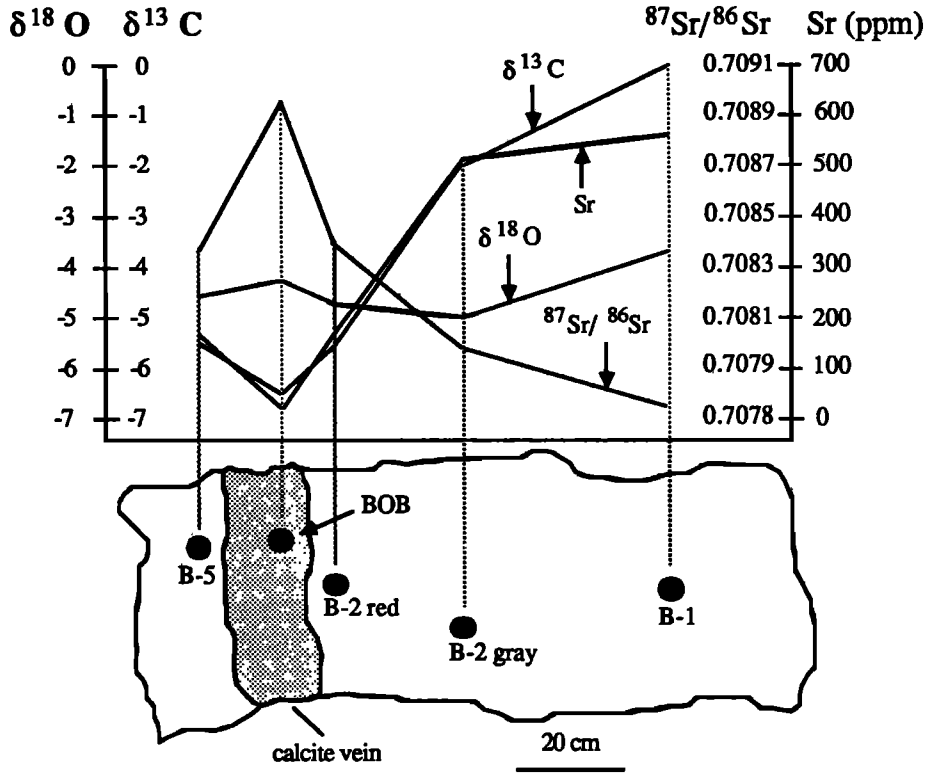


Fig. 8. Sr, $^{87}Sr/^{86}Sr$, and $\delta^{13}C$ variation of limestone near a large calcite vein. The lower Sr concentration, higher $^{87}Sr/^{86}Sr$ ratio, and ligher $\delta^{13}C$ value of the limestone closer to the fracture indicate more intense alteration by vein calcite-precipitating fluids.

aragonitic sediments; this indicates that the Viola has been recrystallized. The $\delta^{18}O$ values of the Viola Limestone are similar to proposed values for calcite precipitated from Ordovician seawater [e.g., Popp et al., 1986], but because the Viola underwent recrystallization, the values were probably reset [Gao et al., 1992]. This recrystallization probably reflects early meteoric diagenesis recognized by Grammer [1985]. The similar isotopic (C, Sr) compositions of the small veins to the host Viola Limestone (Figures 5*a* and 5*c*) suggest that the fluids which precipitated the small veins were buffered by the host limestone [Gao et al., 1992]. The recrystallization of the host Viola Limestone and the precipitation of the small veins was probably caused by ^{18}O depleted meteoric water [Gao et al., 1992].

TOC values from the Viola Limestone indicate the rock is relatively organic-rich. The results of Rock-Eval pyrolysis suggest that the rocks are immature although they are probably just within the oil window. Biomarker analysis on samples from the Viola at Pauls Valley, several miles north of the Arbuckle mountains, show that the oil within the unfractured Viola exhibits the same characteristics as the Viola rock extract [Jones and Philp, 1990]. This suggests that the Viola is one source for the hydrocarbons present in the rock.

The spherical and framboidal magnetic phases found in situ within the Viola are interpreted to be authigenic in origin. The spherical forms are similar to some phases that have been found in magnetic extracts and identified as magnetite [e.g., McCabe and Elmore, 1989]. The partially replaced framboids are also similar to pyrite framboids replaced by magnetite and found in situ in several limestones by Suk et al. [1990*a*, 1990*b*]. On the basis of this similarity, we interpret these phases in the Viola as magnetite.

The results of the fluid inclusion study indicate that the calcites in the large veins precipitated from relatively warm basinal fluids [e.g., London et al., 1990; also manuscript in preparation, 1993]. The presence of base metal minerals (London et al., manuscript in preparation, 1993), high Fe and Mn values, and the radiogenic $^{87}Sr/^{86}Sr$ ratios [Gao et al., 1992] support this interpretation. The oxygen isotopic composition of the calcites further constrains their origin to ^{18}O-enriched fluids [Gao et al., 1992].

The alteration zones around the large veins have much lower Sr concentrations than the host Viola Limestone, which probably reflects additional alteration after early meteoric diagenesis. The similar $\delta^{13}C$ values for the vein calcites and the alteration zones suggest that the fluids that precipitated the calcite in the veins were also responsible for alteration in the surrounding limestone. The $\delta^{13}C$ values, however, could also be related to ^{13}C depleted meteoric water derived from soil zones. The radiogenic $^{87}Sr/^{86}Sr$ values of the limestone in the alteration zones, and the fact that there is more significant alteration closer to the veins, however, suggest that ^{87}Sr-enriched basinal fluids were responsible for the alteration.

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The $\delta^{18}O$ values of the limestone in the alteration zones are similar to the values in the host Viola Limestone. We explain this observation as being due to the nature of the basinal fluids. The $\delta^{18}O$ value depends on the temperature and the $\delta^{18}O$ value of the basinal fluids [Gao et al., 1992]. The similarity of the $\delta^{18}O$ values from the alteration zones and the basal calcite suggest that basinal fluids reset the $\delta^{18}O$ values and reached isotopic equilibrium with the limestone in the alteration zones.

Paleomagnetism of the Viola Limestone

Nonmineralized Zone

Decay of the natural remanent magnetization (NRM) in specimens from the nonmineralized zone occurs during both AF and thermal demagnetization (Figure 9*a*). Thermal demagnetization revealed that the maximum unblocking temperatures were 580°C, although 90% of NRM in most specimens was removed by 450°C. In most specimens, AF demagnetization to 75 mT reduced the NRM by 90%. The results of IRM acquisition and decay measurements corroborate NRM demagnetization results indicating that the magnetization in the specimens is dominated by a low-coercivity phase with maximum unblocking temperatures of 580°C (Figures 10*a* and 10*b*).

After removal of a northerly and steep component at low unblocking temperatures or field strengths, the magnetization isolated in most specimens was of southeast declination and shallow to moderate inclination (Figure 9*a* and Table 2). A fold test, based on six sites from the syncline in the SRS quarry, indicates that the magnetizations from each flank cross during unfolding (Figure 11*a*). An incremental fold test indicates that the best grouping of site directions occurs at 70% unfolding (Figure 11*b*). The mean direction at 70% unfolding is declination 145°E and inclination 19° (k=392, α_{95}=3.4°) and is statistically significant at the 95% level.

Results from the sites in nonmineralized Viola from the south flank of the Arbuckle Anticline are similar to those at the SRS quarry (Table 2). After removal of a northerly and moderate direction at low temperatures, most specimens decayed in a linear direction. The site directions group with a mean of 135/29 before tilt correction and 147/1 after tilt correction (Table 2). The mean at 70% unfolding is 146/10. The magnetization in most specimens from the north flank of the anticline also exhibited linear decay during demagnetization. The in situ mean for these sites is 142/1, whereas the tilt-corrected direction is 115/20.

Interpretation

The results of thermal and AF demagnetization, supported by rock magnetic measurements, suggest that the pervasive magnetization in the nonmineralized Viola is dominated by magnetite. The northerly and steep component removed during initial demagnetization steps is interpreted as modern and viscous in origin. The southeasterly and shallow component is interpreted as the characteristic magnetization. The results from the fold in the SRS quarry suggest this magnetization was acquired during Pennsylvanian folding. The pole position (longitude 127°E, latitude 35°N) for the magnetization falls near the Carboniferous segment of the apparent polar wander path (Figure 12).

The directions from sites on the Arbuckle Anticline are somewhat different from those reported by Peck et al. [1984] from the same sites. For example, the mean in situ direction for the south flank was 126/30 in the former study which compares to 135/29 (Table 2) in the current study. This difference in directions may be due to the incomplete removal of a present-day magnetization by AF demagnetization which was used in the earlier study. Thermal demagnetization used in the current study allowed better isolation of the characteristic magnetization. This interpretation is similar to that suggested by McCabe et al. [1984] in an

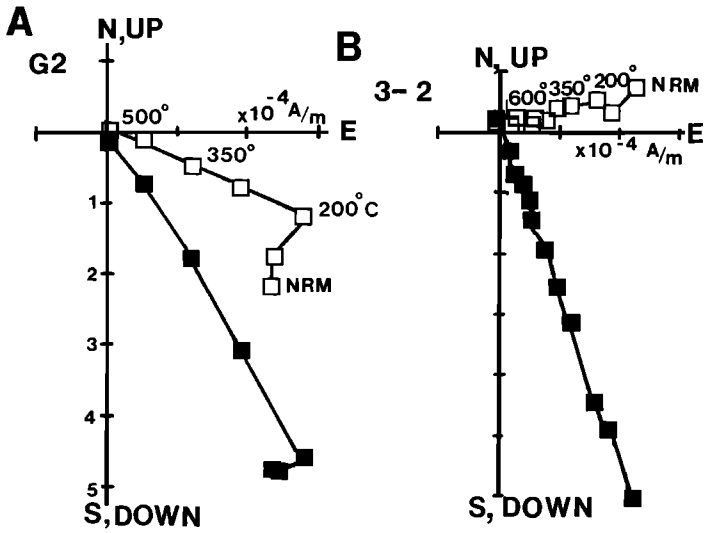


Fig. 9. Representative Zidjderveld diagrams showing decay of the magnetization in both the (*a*) nonmineralized and (*b*) mineralized zones during thermal demagnetization.

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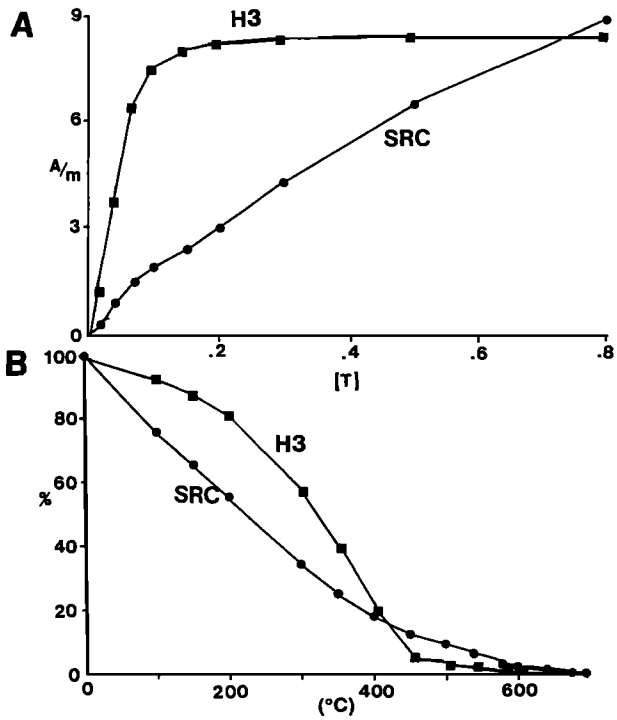


Fig. 10. Representative isothermal remanent magnetization (IRM) acquisition and thermal decay of saturation isothermal remanent magnetization (SIRM) curves for samples of the Viola limestone. (*a*) IRM acquisition curves for limestone specimens in the mineralized (SRC) and nonmineralized (H3) zones. Curve SRC does not achieve saturation by 0.8 T. Note small cusp in curve at 0.1 T, which suggests the presence of a minor amount of a low-coercivity phase. The rise in the curve above 0.2 T indicates the presence of a high-coercivity phase. Curve H3 reaches saturation by 0.2 T, indicating the dominance by a low-coercivity mineral. (*b*) Representative thermal decay of IRM curves for specimens in the mineralized and nonmineralized zones. Continued decay above 580°C in SRC suggests the presence of some hematite. Note maximum unblocking temperatures of 580°C for curve H3, which suggests that the magnetic phase is magnetite.

investigation of the Trenton Limestone.

The SRS quarry and the north flank of the Arbuckle Anticline are between the Washita Valley and Reagan faults, which define a structurally complex area deformed by wrench faulting [Wickham, 1978]. The possibility exists therefore that this faulting could have rotated structural blocks, and magnetizations they contain, around a vertical or near-vertical axis. The results from the SRS locality are similar to the results from the south flank of Arbuckle Anticline where rotation is not likely. This suggests that the magnetizations in the rocks at the SRS quarry have not been rotated around a vertical axis. The magnetizations in rocks from the north flank of the Arbuckle Anticline, however, may have been rotated around a vertical axis. The strike of the beds at the north flank (N60°W) is significantly different than for the SRS locality (N25°W). The rocks hosting the secondary magnetization from the north flank could be rotated or, alternatively, their present orientations could indicate asymmetrical folding.

Low burial temperatures (<100°C) suggest the synfolding magnetization is not thermoviscous in origin but is probably a chemical remanent magnetization (CRM). The presence of authigenic phases interpreted to be magnetite and found in situ is consistent with this interpretation. We note, however, that the phases observed on the SEM are multidomain or pseudo-single-domain in character, and whether they carry the Pennsylvanian CRM is open to question [e.g., Jackson, 1990].

The origin of the CRM is problematic. The geochemical results from the nonmineralized Viola provide no evidence that basinal fluids entered the rocks. Another mechanism, not relying on externally derived fluids, is apparently required to explain the origin of the CRM.

Mineralized Viola

The magnetization in specimens of the vein calcite has high remanent intensities (~10^{-2} A/m) but does not respond in a linear fashion to progressive thermal demagnetization. Thermal demagnetization suggests the presence of two phases with maximum unblocking temperatures of ~100°C and 680°C, respectively.

Paleomagnetic results from the bleached and reddish alteration zones are distinct. Thermal demagnetization of specimens of the reddish limestone next to the mineralized veins reveals maximum unblocking temperatures of 690°C. The magnetization in these specimens does not exhibit significant decay during AF demagnetization. Rock magnetic studies suggest the magnetization is dominated by a high-coercivity phase with maximum unblocking temperatures near 680°C (Figures 10*a* and 10*b*). Consistent south-southeasterly directions were removed during thermal demagnetization of these specimens (Figures 9*b* and 13 and Table 3).

In addition to samples near the veins, samples also were collected along a traverse away from the veins. With increasing distance from a vein, the percent of NRM removed by AF demagnetization increases (Figure 14). Along the same traverse, the results of rock magnetic studies also record a decrease in the ratio of a high-coercivity phase to a low-coercivity phase. Thermal demagnetization of a composite IRM in a specimen collected along the traverse suggests the presence of both hematite and magnetite (Figure 15). The directions in samples collected near but not in direct contact with the veins are also intermediate between the south-southeasterly and shallow high-coercivity component and the southeasterly and steeper low-coercivity component (Table 3).

The NRM intensities in specimens from the bleached altered zones are generally lower than those in the reddish alteration zones or in the nonmineralized zone. The demagnetization trajectories are also not as linear as for the other specimens. The magnetization decays during AF demagnetization, and most maximum unblocking temperatures are near 550°C. Measurements of IRM acquisition and decay suggest the magnetization is dominated by a low-coercivity phase that decays by 580°C, although some specimens contain

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TABLE 2. Summary of Paleomagnetic Data From the Nonmineralized Viola Limestone

Site	N/No	Specimen Directions in Situ Declination, degrees	Inclination, degrees	k	α95	Specimen Directions After Tilt Declination degrees	Inclination degrees	Paleopole Position Latitude, °N	Longitude, °E
				Sooner Rock and Sand Quarry					
G	13/15	140	0	254	2.6	147	15		
N	4/4	139	12	483	4.2	147.2	17		
U	5/5	143	13	288	4.5	151	16		
H	9/11	159	36	680	1.97	134.5	20		
I	7/8	155	32	223	4.0	136	16		
L	6/8	154	26	111	6.4	142	11		
Mean before correction	6/6	148	22	32.6	11.9			35	122
Mean after correction	6/6	145	16	117	6.2			36	127
Mean at 70% correction	6/6	145	19	392	3.4			34.8	126.5
				South Flank Arbuckle Anticline					
VS1A	4/5	135	34	64	11.6	150	4		
VS2A	4/4	138	25	1593	2.3	146	-3		
VS2B	5/5	136	29	250	4.8	147	0		
VS3A	5/5	131	30	435	3.7	146	4		
VS5A	8/8	131	26	421	2.7	142	2		
VS6A	4/5	138	28	148	7.6	147	-3		
Mean of VS sites before tilt	6/6	135	29	402	3.3			24	131
Mean of VS sites after tilt		147	1	402	3.3			43	132
				North Flank of Arbuckle Anticline					
VN1B	3/4	143	7	127	11.0	110	18		
VN2A	3/4	142	-6	104	12.1	120	21		
Mean	2/2	142	1	-	-	115	20		

The value k is a precision parameter and indicates the relative dispersion about the true mean of the magnetic directions in the samples [Fisher, 1953]. The value α95 represents the angular radius about the calculated site mean direction within which there is a 95% probability that the true mean lies. N/No is the ratio of specimens used in statistical calculation to specimens analyzed.

a high-coercivity phase as well. The directions (Table 4) from these specimens are similar to the directions from the nonmineralized Viola as well as samples not in direct contact with the veins (Table 3).

Interpretation

The demagnetization characteristics, supported by the results of the rock magnetic studies, indicate that the calcite in the veins contains both goethite and hematite. This is confirmed by the petrographic studies which indicate the presence of both phases in the rhombic calcite overgrowths.

The pole position (109°E, 51°N; in situ) for the specimens in the reddish alteration zone near the fractures falls on the late Permian segment of the North American apparent polar wander path (Figure 12). The demagnetization results, supported by the rock magnetic and petrographic studies, indicate that the magnetization in these specimens resides in hematite. This magnetization is interpreted as a CRM; the fluids that precipitated the vein calcites were not hot enough to result in a thermoviscous remagnetization. The presence of several forms of authigenic hematite in the altered limestone is consistent with this interpretation.

The trends in the percent AF demagnetization and the rock magnetic results, as well as the change in directions, suggest that with increasing distance from the veins the hematite magnetization is reduced whereas the magnetite component from the nonmineralized Viola becomes more prevalent. This indicates that there is a alteration halo with a Permian CRM surrounding the fractures. This change in magnetic mineralogy and direction away from the veins coincides with the trends noted in the geochemical results (Figure 16). On the basis of both the geochemical trends and rock magnetic observations, we suggest the Permian CRM in the alteration zone was acquired in response to fluid migration in the veins. The fluid or

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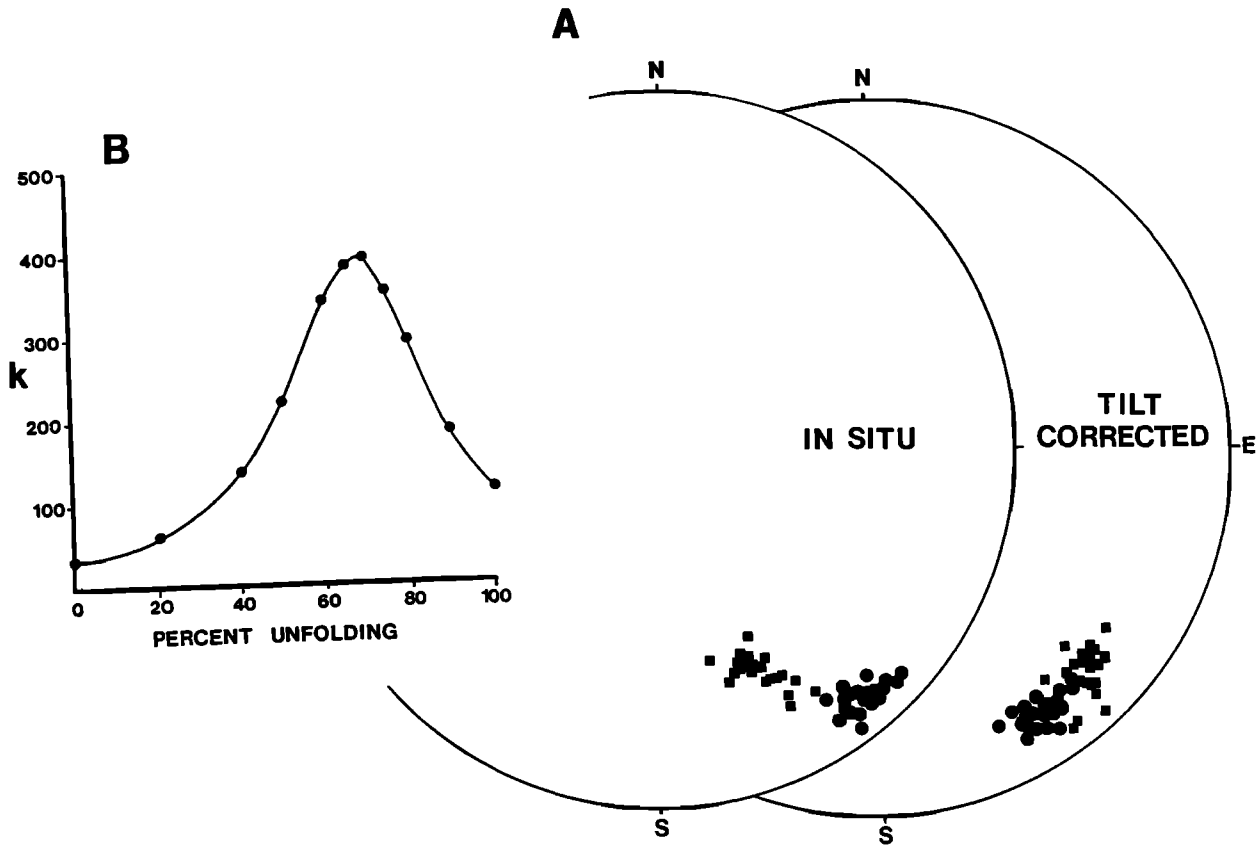


Fig. 11. (*a*) Schmidt equal-area projection of specimens from six sites in the nonmineralized zone, before and after tilt correction. Square symbols represent specimens from eastward dipping limb; circles represent specimens from westward dipping limb (all specimens plot on lower hemisphere). Note that the directions cross during complete tilt correction, indicating a synfolding magnetization. (*b*) An incremental fold test was performed on the site means from six sample sites from the nonmineralized zone, and it was determined that the best grouping (k) of specimens came at 70% unfolding, indicative of a synfolding magnetization.

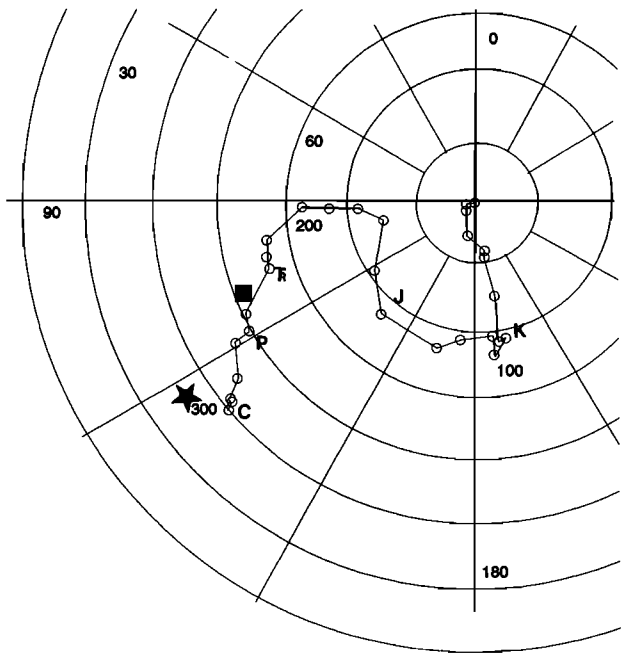


Fig. 12. North American apparent polar wander path [Irving and Irving, 1982] with the paleopoles calculated from the mineralized zone (square) and the nonmineralized zone (star). Open circles correspond to 10 Ma means from Irving and Irving [1982].

fluids apparently caused hematite to precipitate and/or to replace pyrite in most of the alteration zones.

The results from the bleached alteration zones around some of the hydrocarbon-bearing veins suggest a magnetization residing in magnetite with a direction similar to that in the nonmineralized limestone. Although these limestones are altered on the basis of geochemical results, the remagnetization halo is largely absent.

Discussion

The results of this study have implications for dating mineralization associated with basinal fluids and for the origin of pervasive CRMs that reside in magnetite.

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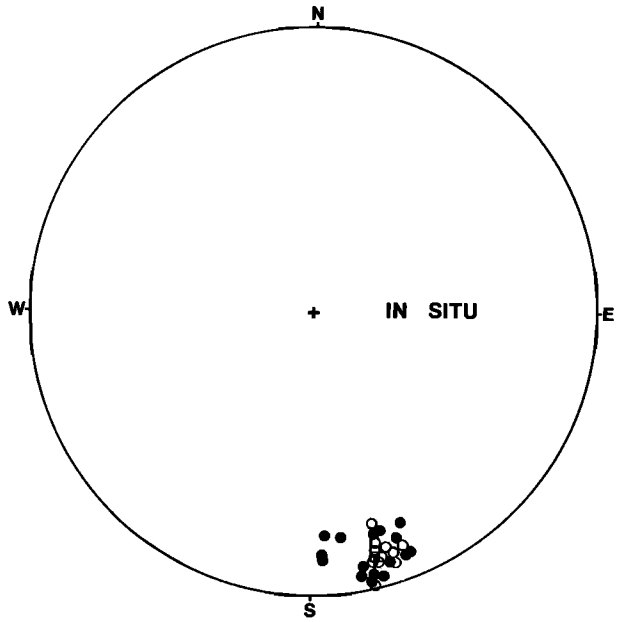


Fig. 13. Schmidt equal-area projection of directions of magnetization (in situ) for specimens in the mineralized zone in direct contact with a calcite-filled fracture. Solid symbols represent projections in the lower hemisphere, open symbols represent projections in the upper hemisphere.

Dating of Mineralization

A number of paleomagnetic studies have attempted to date Mississippi-Valley-type (MVT) mineralization. Most of these studies report a late Paleozoic magnetization residing in magnetite [e.g., Beales et al., 1974; Wu and Beales, 1981], although it has been difficult to establish a direct relationship between the magnetization and mineralization. Wisniowiecki et al.[1983] argue that mineralization in the Viburnum Trend postdated magnetization acquisition whereas Bachtadse et al. [1987] suggest the magnetization in the east Tennessee MVT district is post-ore in age. In a recent study, Pan et al. [1990] argue that mineralization in the northern Arkansas and Tri State districts may be coeval with the acquisition of a Permian magnetization in magnetite, although they point out the connection is not conclusive. The character of the magnetizations described in these studies is similar to that of the pervasive remagnetization in the Viola. Farr and Gose [1991], however, report a Permian magnetization in hematite from a MVT deposit in central Texas that predates mineralization.

The geochemical results from the alteration zones provide clear evidence that the basinal fluids altered the host Viola Limestone around the veins. The paleomagnetic results also indicate a remagnetization halo around most of the veins. The coincidence of the geochemical alteration with the remagnetization halo suggests that migration of basinal fluids caused the Permian CRM.

The bleached alteration zones around some veins with asphalt do not contain the remagnetization halo. The chemical conditions in the bleached alteration zones were apparently not right for the basinal fluids to cause the Permian magnetization. The reason for this is unclear, although it may be related to the presence of hydrocarbons and the nature of the rhombic overgrowths. The rhombic overgrowths in these veins do not contain inclusions of oxides in contrast to the overgrowths in veins with the remagnetization halo. This suggests that the fluid which precipitated the rhombic overgrowths with oxide inclusions could be responsible for the remagnetization. This fluid had the appropriate chemistry during its evolution to cause hematite authigenesis and acquisition of the CRM.

The calcites precipitated in the veins record a complex history of changing chemical conditions. As a result, specifically relating the CRM to one of the two mineralizing fluids is not straightforward. For example, the fluid that

TABLE 3. Summary of Paleomagnetic Data for the Mineralized Zone in the Ordovician Viola Limestone

Site	N/No	Mean Direction (in Situ) Declination degrees	Inclination degrees	k	α95	Paleopole Position Latitude, °N	Longitude, °E
		Limestone Specimens in Direct Contact With Fractures					
3	4/4	164	-0.2	115	8.6	52.6	109.7
C	9/9	163	6.9	33	9.1	49.1	108.5
D	4/6	165	-3.7	99	9.3	54.6	109.4
F	6/7	168	12.5	78	7.6	47.7	100.6
P	6/6	163	-3.9	53	9.3	54.1	112
Q	5/5	164	-10.7	20	17.3	57.6	112.6
Mean of sites	6/6	165	0.1	91	7.1	52.7	108.6
		Limestone Specimens not in Direct Contact With Fractures					
B	5/6	158	12.3	119	7.0	44.4	114
O	7/7	160	20.4	294	3.5	41.2	109.5
A	6/8	159	13	325	3.7	44.4	113.1

See note for Table 2.

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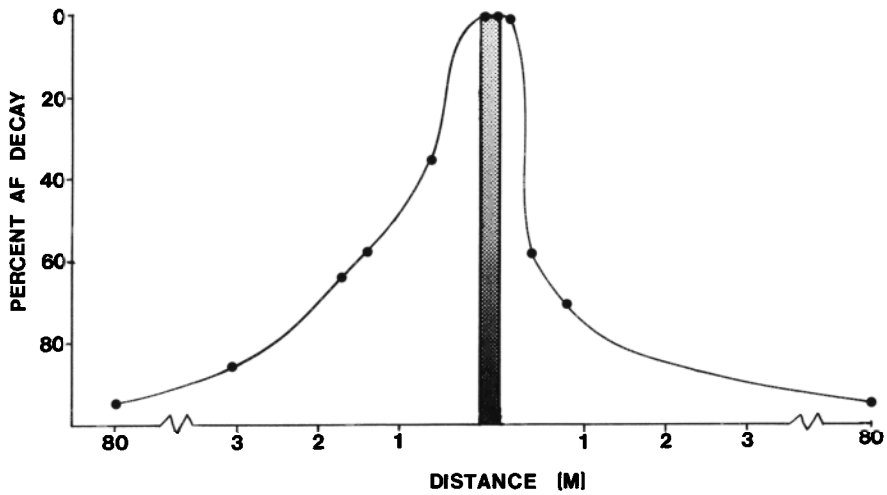


Fig. 14. Summary diagram showing percent of total magnetization removed by alternating field (AF) decay as a function of distance from a calcite-filled fracture. Note that the highest- coercivity (low AF decay) specimens are located within or very close to the fracture, but as distance increases, a low-coercivity (high percent AF decay) phase begins to emerge.

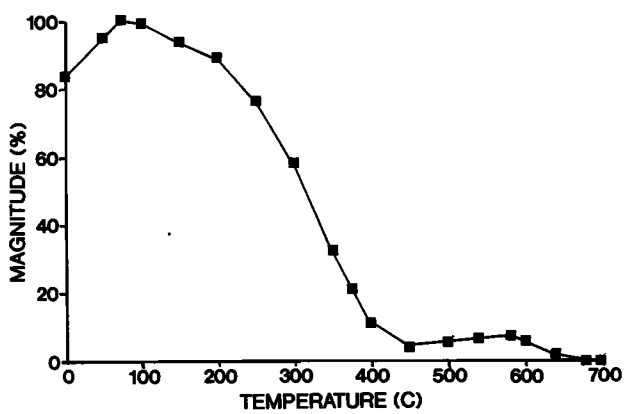


Fig. 15. Representative thermal decay of composite isothermal remanent magnetization curve for a specimen from the mineralized zone not in direct contact with a fracture. Note the presence of two magnetic phases; one component is removed by 580°C, while the second decays to 680°C.

deposited the basal calcite cannot be ruled out as the agent of hematite authigenesis and remagnetization. The $\delta^{13}C$ values of the altered limestone are closer to the $\delta^{13}C$ values of the basal calcite than to the rhombic overgrowths [Gao et al., 1992], which suggests that the fluid which precipitated the basal calcite caused the geochemical alteration and perhaps the Permian CRM. Unfortunately, there are no firm constraints on the oxidation state of the fluid from which the basal scalenohedral calcite was deposited (Type I inclusions, Table 1). If the Type III inclusions (Table 1) are correctly identified as mixtures of Type I and Type II fluids, however, we can at least say that one or both of these fluid components, Type I or Type II, was fairly oxidizing (on the hematite-pyrite buffer, on the basis of pyrite + goethite associated with the fluid inclusions).

The presence of a few inclusions with meteoric fluids along healed fractures also raises the question as to whether fresh waters could have caused hematite precipitation and the associated CRM. These inclusions, however, are not common. In addition, if they came from the surface, which is likely, or were buffered by the host Viola limestone, then they would have changed the Sr isotope ratios in the alteration zone, which is not observed.

Although it is not possible to conclusively relate the CRM to one of the two mineralizing fluids, a strong case can be made that the basinal fluids were responsible for the hematite authigenesis and the associated remagnetization. The pole position for the CRM suggests that the timing of the mineralization can be constrained to the Late Permian. The occurrence of mineralization in well cuttings from the fault

TABLE 4. Paleomagnetic Results From Bleached Alteration Zones

Site	N/No	Mean Direction (in Situ) Declination degrees	Inclination degrees	k	α95	Mean Tilt-Corrected Direction Declination degrees	Inclination degrees
HV1	5/8	151	29	33	13.5	142	17
HV2	4/4	153	24	66	11.4	144	13

See note for Table 2.

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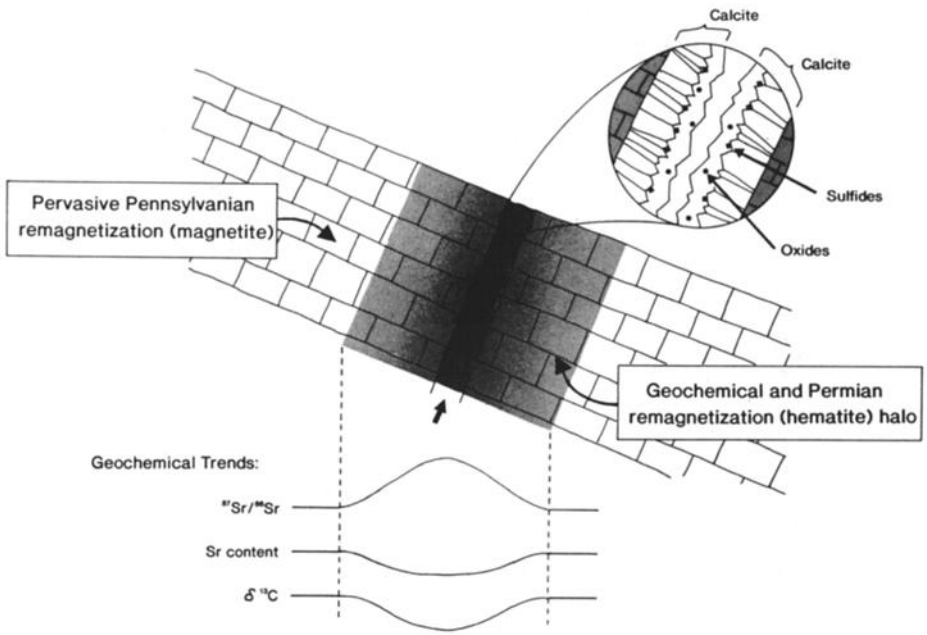


Fig. 16. Schematic model that illustrates a vein with the geochemical and remagnetization halos.

zone suggests the fluids moved up along a permeable fault zone. This is consistent with recently proposed models that employ focused flow for the origin of some MVT deposits [e.g., Spirakis and Heyl, 1988; Clendenin and Daune, 1990].

The mineralization found in the Viola may be associated with the MVT mineralization at the Davis Zinc Field. Estimates of fluid inclusion microthermometry by Wheeler [1989] from a deposit at the Davis Zinc Field suggest dolomite + sphalerite precipitation from warm (100°C to 130°C) and saline brines (21 wt % NaCl equivalent). Authigenic feldspars intergrown with sphalerite formed in the Late Pennsylvanian-Permian (251-285 Ma) based on K-Ar dating [Wheeler,1989]. Wheeler [1989] suggested that the feldspars were temporally related to the MVT mineralization as cogenetic or at least before sphalerite formation.

The temperatures of formation for the calcites in the veins at the SRS quarry are lower than for the dolomites at the Davis Zinc deposit. The remagnetization is also apparently somewhat younger than the date for the feldspars, although there is overlap with one of the feldspar samples which gave a date of 251±24 [Wheeler, 1989]. The mineralization in the veins at the SRS quarry could represent an updip, cooler, and late-stage phase of the mineralization at the Davis Zinc Field.

Origin of the Pervasive Remagnetization in Magnetite

Pervasive secondary magnetizations residing in magnetite that are similar to the pervasive CRM in the Viola are common in carbonate rocks [McCabe and Elmore, 1989]. A pervasive late Paleozoic CRM that resides in magnetite and was acquired during deformation also occurs in the laterally equivalent Bigfork Chert along Black Knob Ridge in the frontal Ouachita Mountains [Hillegeist et al., 1992]. The origin of pervasive secondary magnetizations that reside in magnetite, however, is controversial [e.g., McCabe and Elmore, 1989]. The debate has generally focused on two mechanisms: chemical and thermoviscous processes. Although some of these pervasive secondary magnetizations are thermoviscous in origin [e.g., Kent, 1985], low burial temperatures, and the presence of authigenic magnetite, have led many workers to suggest that many secondary magnetizations are chemical in origin [e.g., McCabe and Elmore, 1989]. Migration of fluids is currently a popular hypothesis invoked to explain the origin of many secondary magnetizations [e.g., McCabe et al., 1989; Lu et al., 1990; Pan et al., 1990] as well as other events that occurred in the late Paleozoic [e.g., Oliver, 1986, 1992].

The geochemical results from the nonmineralized Viola, particularly the $^{87}Sr/^{86}Sr$ ratios, provide no evidence that basinal or other exotic, externally derived fluids entered the Viola except locally around veins. The lack of geochemical evidence for basinal fluids suggests that another fluid or another mechanism is needed to explain the origin of the pervasive CRM in the Viola Limestone. This interpretation is consistent with a recent modeling study which questions the viability of mechanisms for pervasive flow of brines [Deming and Nunn, 1991].

Thermoviscous mechanisms can explain some pervasive secondary magnetizations, but burial temperatures in the Viola (< 100°C) are too low to reset a magnetization based on theory for single-domian magnetite [e.g., Pullaiah et al., 1975; Worm and Jackson, 1988; Jackson, 1990]. The burial temperatures are also too low to reset a magnetization based on experimental results which indicate that thermoviscous remagnetization can extend to higher demagnetization temperatures than predicted by single-domain theory [e.g., Kent, 1985].

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Strain-related processes have been invoked to explain some secondary magnetizations [e.g., Hudson et al., 1989], but these mechanisms are considered unlikely for the Viola because the magnetization is regionally pervasive and occurs in areas that have different strain histories. We cannot, however, conclusively rule out the possibility that strain-related processes have affected the pervasive magnetization.

Chemical processes unrelated to exotic externally derived fluids are an obvious alternative for the origin of the pervasive magnetization. On the basis of the chemical data, the only fluids to pervasively enter the Viola are interpreted to be meteoric in nature. These meteoric fluids, in combination with other diagenetic processes that occurred in the Viola, could explain magnetite authigenesis. For example, magnetite authigenesis could be related to maturation or diagenesis of organic matter in the Viola Limestone. Hydrocarbons have been related to precipitation of authigenic magnetite [Elmore et al., 1987; McCabe et al., 1987] and also to acquisition of a chemical magnetization [Elmore and Crawford, 1990]. Another possible mechanism that could be related to magnetite authigenesis is the transformation of smectite to illite, which releases iron [e.g., Boles and Franks, 1979]. These and possibly other chemical processes mediated by relatively low burial temperatures could explain the pervasive CRM in the Viola as well as other pervasive secondary magnetizations in carbonate rocks.

Conclusions

In summary, the results of this study suggest that proposed connections between pervasive CRMs and pervasive flow of basinal fluids should be viewed with caution. There is no geochemical evidence for basinal fluids in Viola Limestone samples that contain the pervasive CRM in magnetite. Geochemical evidence for basinal fluids is found only around veins which were the conduits for focused flow. A chemical process, perhaps mediated by low burial temperatures, is needed to explain the pervasive CRM. Although we cannot conclusively identify the mechanism which caused the pervasive CRM in the Viola, it is clear that tests for mechanisms other than basinal fluids are needed.

Altered limestone around the mineralized fractures in the Viola Formation contains a Permian CRM residing in hematite. Geochemical data indicate that this alteration zone was caused by basinal fluids. The coincidence of the geochemical and remagnetization trends suggest that basinal fluid migration led to CRM acquisition. The paleomagnetic and geochemical results therefore suggest the mineralization occurred in the Permian.

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